Extremal Connected Still Lifes in Square Boards

Siddhartha Mahajan

Lossfunk Research Laboratory, Bengaluru, Karnataka, India

Corresponding author: siddharthamahajan03@gmail.com

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Abstract

Let $C(n)$ denote the largest population of a stable Conway Life pattern in an $n \times n$ square whose live cells form one component under Moore adjacency; the exterior of the square is dead. We determine

$$C(1), \ldots, C(25) = 0, 4, 6, 8, 11, 18, 28, 32, 43, 53, 63, 76, 88,$$
$$102, 119, 134, 151, 170, 188, 210, 230, 251, 276, 299, 325$$

and the further values $C(29) = 435$, $C(31) = 497$, $C(32) = 528$, $C(33) = 562$, $C(37) = 704$, $C(39) = 782$, $C(41) = 863$, $C(45) = 1037$, $C(49) = 1228$, $C(53) = 1434$, $C(57) = 1657$, and $C(65) = 2150$. The first unresolved case satisfies $350 \le C(26) \le 352$, and every unresolved size $n \le 50$ has a gap of at most five cells.

A live 2×2 block is an isolated Moore component, so every connected still life with more than four cells is block-free. Combining this local obstruction with the Chu–Stuckey wastage identity yields finite-state boundary relaxations, and exact min-plus composition of edge automata and corner relations gives, for every $n \ge 14$,

$$C(n) \le \min\left\{M(n), \left\lfloor \frac{4n^2 + 5n - d_{n \bmod 8}}{8} \right\rfloor\right\}, \qquad (d_0, \ldots, d_7) = (26, 23, 18, 23, 20, 23, 26, 11),$$

where $M(n)$ is the unrestricted maximum. Explicit period-eight band families give $C(n) \ge n(n+1)/2 - 4$ for every $n \ge 1$, with equality of the constant at $n = 5$ and $n = 8$, and hence $\lim_{n\to\infty} C(n)/n^2 = 1/2$.

1 Introduction

A still life is a fixed point of Conway's Game of Life: each live cell has two or three live neighbours, and each dead cell has a number of live neighbours different from three. In the finite maximum-density problem the live set is confined to an $n \times n$ square and every cell outside the square is dead. Exact methods for this problem include variable elimination and boundary relaxations [2, 3], and the unrestricted optimum, denoted $M(n)$ below, was completely determined by Chu and Stuckey [4].

The connected problem, posed as a benchmark of the 2013 Answer Set Programming Competition [5], adds a global condition: the live cells must form one component under horizontal, vertical, and diagonal adjacency. This is not a small perturbation. Dense unrestricted optima often consist of a large maze and several stable 2×2 blocks, and such a block cannot be attached to any other live cell, so it is forbidden in every connected pattern of population greater than four. The problem was also one of the challenges of the Verifiable Problems Track of CAISc 2026 [9], posed for the thirteen sizes $8, 9, 10, 11, 13, 15, 16, 17, 19, 20, 21, 31, 32$. There, an accepted artifact is a board checked by a deterministic verifier, that is, a lower bound. The seven accepted submissions on this problem [10, 11, 12, 13, 14, 15, 16] and the ideas taken from them are summarized in appendix A. All thirteen challenge sizes are now determined exactly.

Every lower bound in this paper is an explicitly verified board. Upper bounds come from proof-logged threshold instances, exhaustive finite-state transfers, or min-plus boundary certificates; no exact value is inferred from an unsuccessful search.

Theorem 1.1 (Main results)**.** *Let $C(n)$ be the connected finite-board optimum of definition 2.2. Then:*

(i)
$$C(1), \dots, C(25) = 0, 4, 6, 8, 11, 18, 28, 32, 43, 53, 63, 76, 88,$$
$$102, 119, 134, 151, 170, 188, 210, 230, 251, 276, 299, 325.$$

(ii) *Moreover*
$$C(29) = 435, \quad C(31) = 497, \quad C(32) = 528, \quad C(33) = 562,$$
$$C(37) = 704, \quad C(39) = 782, \quad C(41) = 863, \quad C(45) = 1037,$$
$$C(49) = 1228, \quad C(53) = 1434, \quad C(57) = 1657, \quad C(65) = 2150.$$

(iii) *For the sixteen remaining sizes $n \le 50$, namely $n = 26, 27, 28, 30, 34, 35, 36, 38, 40, 42, 43, 44, 46, 47, 48, 50$, the optimum lies in the interval of table 2. In particular $350 \le C(26) \le 352$, and every interval has width at most five.*

(iv) *For every $n \ge 14$,*
$$C(n) \le \min\{M(n), U(n)\}, \qquad U(n) = \left\lfloor \frac{4n^2 + 5n - d_{n \bmod 8}}{8} \right\rfloor, \tag{1}$$
where
$$(d_0, d_1, d_2, d_3, d_4, d_5, d_6, d_7) = (26, 23, 18, 23, 20, 23, 26, 11). \tag{2}$$

(v) *For every $n \ge 1$,*
$$C(n) \ge \frac{n(n+1)}{2} - 4, \tag{3}$$
and the constant 4 cannot be lowered: equality holds at $n = 5$ and $n = 8$. Hence $\lim_{n\to\infty} C(n)/n^2 = 1/2$.

The model and the two local reductions are set up in section 2. Section 3 proves the upper bound (iv) by boundary automata. Section 4 closes the exact values (i) and (ii), section 5 records the ledger (iii), and section 6 proves (v). Section 7 lists the remaining questions. The appendices contain the CAISc provenance, the details of the finite closures, the repair methods, earlier construction families, the computational record, and an atlas of all verified boards.

2 Finite-board model and local structure

For $n \ge 1$ let $\mathcal{B}_n = \{0, 1, \dots, n-1\}^2 \subset \mathbb{Z}^2$, and let $\mathcal{N}(z) = \{z + (a, b) : (a, b) \in \{-1, 0, 1\}^2 \setminus \{(0, 0)\}\}$ be the Moore neighbourhood of z. For $S \subseteq \mathcal{B}_n$ put $\nu_S(z) = |S \cap \mathcal{N}(z)|$ and extend the indicator of S by zero to all of $\mathbb{Z}^2$.

Definition 2.1. *A set $S \subseteq \mathcal{B}_n$ is a finite-board still life if, for every $z \in \mathbb{Z}^2$, $z \in S$ implies $\nu_S(z) \in \{2, 3\}$ and $z \notin S$ implies $\nu_S(z) \ne 3$. It is connected if the graph on S with edges between Moore-adjacent cells is connected.*

Definition 2.2. *For $n \ge 2$, let*
$$C(n) = \max\{|S| : S \subseteq \mathcal{B}_n \text{ is a nonempty connected still life}\},$$
and set $C(1) = 0$, since a single live cell is unstable. The unrestricted optimum $M(n)$ is defined by omitting connectivity.

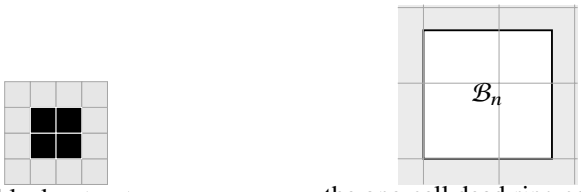
Lemma 2.3 (Complete exterior ring)**.** *Let $R_n = \{-1, 0, \dots, n\}^2 \setminus \mathcal{B}_n$. A set $S \subseteq \mathcal{B}_n$ is a still life if and only if the Life conditions hold at every cell of $\mathcal{B}_n \cup R_n$.*

Proof. Every point outside $\mathcal{B}_n \cup R_n$ has Chebyshev distance at least two from $\mathcal{B}_n$, hence has no live neighbour and cannot be born. □

Lemma 2.4 (Sealed block)**.** *If a stable Life pattern contains all four cells of a unit 2×2 square, then those four cells form an entire Moore-connected component.*

Proof. Each block cell already has the other three as live neighbours. An additional live cell Moore-adjacent to the block would give some block cell at least four live neighbours. □

Corollary 2.5 (Block-free relaxation)**.** *Every connected still life with more than four live cells contains no live 2×2 block.*



a live block saturates all four survival counts

the one-cell dead ring contains every exterior birth constraint

Figure 1: The two local reductions. The gray cells around the live block cannot be live; the shaded layer is the complete exterior verification ring.

2.1 The unrestricted benchmark and wastage

Theorem 2.6 (Chu–Stuckey)**.** *Every connected still life is unrestricted, so $C(n) \le M(n)$. The values of $M(n)$ for $n \le 60$ are given by the exact table in [4]; those used in the ledger appear in table 2. For $n \ge 61$, let*

$$\mathcal{R}_{-2} = \{0, 1, 3, 8, 9, 11, 16, 17, 19, 25, 27, 31, 33, 39, 41, 47, 49\} \pmod{54}.$$

Then

$$M(n) = \begin{cases} \left\lfloor \dfrac{n^2}{2} + \dfrac{17n}{27} - 2 \right\rfloor, & n \bmod 54 \in \mathcal{R}_{-2}, \\ \left\lfloor \dfrac{n^2}{2} + \dfrac{17n}{27} - 1 \right\rfloor, & n \bmod 54 \notin \mathcal{R}_{-2}. \end{cases} \tag{4}$$

Chu and Stuckey redistribute two tokens initially placed at every lattice cell. Each live cell receives at least four tokens, and the undistributed tokens define a nonnegative local wastage. For a dead centre with 0, 1, 2, 3, 4 live orthogonal neighbours the preliminary charge is 2, 1, 0, 0, 2; a live centre is charged for eligible dead orthogonal donors beyond two, with the four exceptional donor and recipient patterns of [4] assigned charge zero. On a finite dead-exterior square the identity is exact.

Theorem 2.7 (Wastage identity)**.** *For every finite-board still life $S \subseteq \mathcal{B}_n$ there is a sum $W(S)$ of nonnegative radius-one local charges and side contributions such that*

$$W(S) = 2n^2 + 4n - 4|S|, \qquad |S| = \frac{2n^2 + 4n - W(S)}{4}. \tag{5}$$

In particular $W(S) \equiv 2n^2 + 4n \pmod 4$.

Lemma 2.8 (Partial-charge upper bound)**.** *If a set of local terms completely determined by a boundary annulus has total charge at least B, then every full still-life completion satisfies $|S| \le \lfloor (2n^2 + 4n - B)/4 \rfloor$.*

Proof. All omitted wastage terms are nonnegative, so $W(S) \ge B$; substitute in (5). □

3 Boundary automata and a residue-eight upper bound

The connected upper bound is a relaxation. By corollary 2.5 every relevant connected pattern is block-free, so we minimize wastage over all locally stable, block-free boundary annuli, without requiring that the annulus extend to a connected interior. The relaxed class is larger, so its minimum charge is a valid lower bound on the charge of every connected board.

3.1 Straight edges, corners, and min-plus composition

Fix a depth $h \ge 3$. A straight-edge state is an ordered pair (a, b) of binary column masks of height $h+2$: the outermost bit is the dead exterior, the next h bits are the charged boundary strip, and the last bit is an inward halo. A transition $(a, b) \to (b, c)$ is admitted exactly when appending c satisfies the Life rule at every centre whose neighbourhood

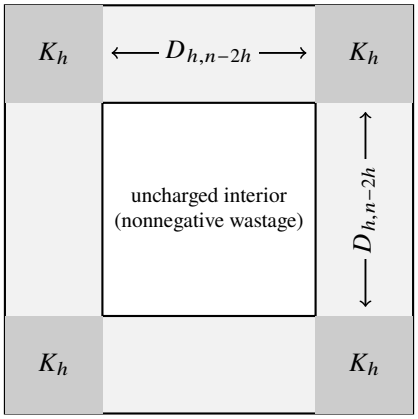


Figure 2: The square boundary as a cyclic min-plus product of four straight path matrices and four corner relations.

has just become complete, suppresses births at the exterior centre finalized by the transition, and creates no live 2×2 block. Its weight is the sum of the newly finalized local wastage terms. With E_h the weighted adjacency matrix and $(A \otimes B)_{ij} = \min_k (A_{ik} + B_{kj})$ the min-plus product, the minimum-cost length-L path matrix satisfies

$$D_{h,L+1} = D_{h,L} \otimes E_h. \tag{6}$$

A corner is not determined by two independent edge endpoints, so the complete $(h+1) \times (h+1)$ corner patches are enumerated directly, giving a weighted corner relation K_h. A depth-h ring has straight length $n - 2h$ on each side and minimum charge

$$B_h(n) = \min_{\alpha_j, \beta_j} \sum_{j=0}^{3} \left(D_{h,n-2h}(\alpha_j, \beta_j) + K_h(\beta_j, \alpha_{j+1}) \right), \qquad \alpha_4 = \alpha_0. \tag{7}$$

Straight strips and corners partition the annular local terms, every full board projects to a feasible ring, and the uncharged interior terms are nonnegative, so by lemma 2.8

$$C(n) \le \left\lfloor \frac{2n^2 + 4n - B_h(n)}{4} \right\rfloor. \tag{8}$$

The finite certificates rest on an entrywise matrix recurrence, not on a recurrence for the scalar minimum.

Lemma 3.1 (Propagation of a min-plus period)**.** *If $D_{h,L_0+p} = D_{h,L_0} + \delta$ entrywise, including equality of the $+\infty$ pattern, then $D_{h,L+p} = D_{h,L} + \delta$ for all $L \ge L_0$.*

Proof. By (6), $D_{h,L_0+p+1} = (D_{h,L_0} + \delta) \otimes E_h = D_{h,L_0+1} + \delta$, and the argument iterates; equality of the infinite entries preserves reachability. □

Since the corner relation is fixed, an entrywise side recurrence passes through (7): increasing n by eight increases the four-side minimum by $4 \cdot 3 = 12$.

3.2 The bound

At depth three the block-free edge graph has 214 states and 771 arcs, and an integer potential p on its states satisfies $8w(e) - 3 + p(s) - p(t) \ge 0$ for every arc $e : s \to t$ of wastage $w(e)$; summing around a cycle shows that every cycle has mean weight at least $3/8$, and length-eight cycles of cost three attain it. Together with the complete 4×4 corner census and the entrywise recurrence $D_{3,L+8} = D_{3,L} + 3$ for $L \ge 20$, this gives the depth-three bound. Depth four retains one more inward layer. The certificate data for depths three through six are listed in appendix B.

Proposition 3.2 (Depths three and four)**.** *For $n \ge 10$ and $n \ge 14$ respectively,*

$$C(n) \le U_3(n) = \left\lfloor \frac{4n^2 + 5n - b_{n \bmod 8}}{8} \right\rfloor, \qquad C(n) \le U_4(n) = \left\lfloor \frac{4n^2 + 5n - c_{n \bmod 8}}{8} \right\rfloor,$$

where $(b_0, \ldots, b_7) = (18, 15, 10, 15, 12, 15, 18, 3)$ and $(c_0, \ldots, c_7) = (18, 23, 18, 23, 20, 15, 26, 11)$.

Residues zero and five require equality-language arguments. At depth five the minima at $n = 21, 29, 37, 45, 53$ are one unit above the depth-four charges, and the restricted matrix recurrence propagates the improvement. For residue zero, the equality rings of depth five lie in a small support, and a safe depth-six lift of that support, which keeps every valid incoming and outgoing edge and both choices of the new halo bit, contains no ring at the old target for $n = 16, 24, 32, 40$; the restricted matrix again increases entrywise by three.

Proposition 3.3 (Residues five and zero)**.** *If $n \geq 21$ and $n \equiv 5 \pmod 8$, then $C(n) \leq U_4(n) - 1$. If $n \geq 16$ and $n \equiv 0 \pmod 8$, then $C(n) \leq U_3(n) - 1$.*

Theorem 3.4 (Connected boundary upper bound)**.** *For every $n \geq 14$, $C(n) \leq U(n)$ with U as in (1) and d as in (2). Therefore $C(n) \leq \min\{M(n), U(n)\}$.*

Proof. Use U_4 in residues $1, 2, 3, 4, 6, 7$. In residue five, subtracting one from U_4 changes $c_5 = 15$ to $d_5 = 23$; in residue zero, subtracting one from U_3 changes $b_0 = 18$ to $d_0 = 26$. The unrestricted inequality is theorem 2.6. □

Corollary 3.5 (Upper-bound recurrence)**.** *For every $n \geq 14$, $U(n + 8) = U(n) + 8n + 37$.*

Proof. The residue correction is unchanged, and $4(n + 8)^2 + 5(n + 8) - (4n^2 + 5n) = 8(8n + 37)$ is divisible by eight, so the increment passes through the floor. □

This is a recurrence for a proved upper bound, not for $C(n)$: boundary feasibility does not guarantee a connected interior completion.

4 Exact values

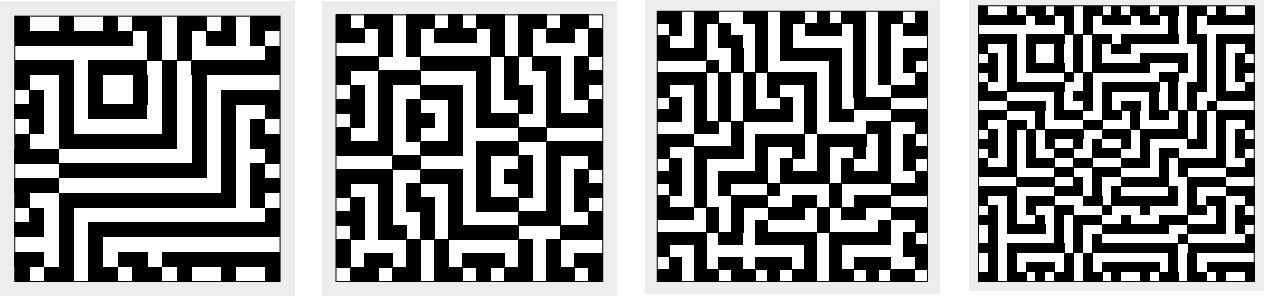
The exact values of theorem 1.1 are closed by the arguments of table 1. In every row the lower bound is a verified connected board. The proof index of the accompanying repository [8] records, for some small sizes, further independent routes: equality with $M(n)$ at $n = 2, 3, 4, 6, 7, 9$ and complete block-free row enumerations at $n = 5, 8, 10, 11$.

Table 1: Upper-bound method for each exact value.

sizes	exact values	upper-bound mechanism
1	0	a single live cell dies
2–11	4, 6, 8, 11, 18, 28, 32, 43, 53, 63	complete connected threshold encodings with proof-logged refutations (appendix C.1)
12, 15, 20, 23, 31, 39	76, 119, 210, 276, 497, 782	connected boards attain $M(n)$
13	88	exhaustive block-free row dynamic program (appendix C.2)
14	102	complete depth-four low-wastage belt enumeration (appendix C.3)
16, 21, 24, 25, 29, 32, 33	134, 230, 299, 325, 435, 528, 562	connected boards attain $U(n)$
37, 41, 45, 49, 53, 57, 65	704, 863, 1037, 1228, 1434, 1657, 2150	connected boards attain $U(n)$
17	151	depth-five equality support and zero-charge depth-six lift (appendix C.4)
18	170	depth-seven exclusion of population 171 (appendix C.5)
19	188	depth-nine exclusion of population 189 (appendix C.6)
22	251	depth-eleven exclusion of population 252 (appendix C.7)

4.1 Closures by the unrestricted optimum and by the boundary bound

At $n = 12, 15, 20, 23, 31, 39$ connected boards attain $M(n) = 76, 119, 210, 276, 497, 782$, so $C(n) = M(n)$ there. At $n = 23, 31, 39$ the boundary formula is one cell weaker, since $U(23) = 277$, $U(31) = 498$, and $U(39) = 783$. Elsewhere the boundary bound is attained.



$n = 18, 170$ $n = 19, 188$ $n = 22, 251$ $n = 29, 435$

Figure 3: Exact constructions at $n = 18, 19, 22, 29$ with their populations. The first three are optimal by the specialized closures, and $C(29) = U(29)$ by proposition 4.1.

Proposition 4.1 (Closure by the boundary bound)**.** *For $n \in \{16, 21, 24, 25, 29, 32, 33, 37, 41, 45, 49, 53, 57, 65\}$ there is a verified connected still life in $\mathcal{B}_n$ of population $U(n)$, so $C(n) = U(n)$:*

n	16	21	24	25	29	32	33	37	41	45	49	53	57	65
$C(n)$	134	230	299	325	435	528	562	704	863	1037	1228	1434	1657	2150

Proof. By theorem 3.4, $C(n) \le U(n)$ for $n \ge 14$, and each listed board has exactly $U(n)$ live cells, is stable on $\mathcal{B}_n \cup R_n$, and has one Moore component. These properties are recomputed by the audit script of appendix F. □

The exact values with $n \equiv 5 \pmod 8$, $21 \le n \le 53$, and with $n \equiv 1 \pmod 8$, $25 \le n \le 65$, form the chains

$$21 \to 29 \to 37 \to 45 \to 53 \qquad \text{and} \qquad 25 \to 33 \to 41 \to 49 \to 57 \to 65,$$

in which consecutive values differ by exactly $8n+37$, the increment of U in corollary 3.5; for example $C(65) = 2150 = 1657 + 8 \cdot 57 + 37$. These are finite verified chains, not a proof that they continue.

4.2 Specialized closures

The remaining exact values need finite arguments beyond theorem 3.4. We summarize them here; the complete arguments are in appendix C.

$n \le 11$. Complete threshold encodings with a rooted spanning-tree connectivity model; unsatisfiability at $C(n) + 1$ is certified by replayed proofs.

$n = 13$. Any connected board with 89 cells is block-free; an exhaustive block-free row dynamic program, over 3,926,925,326 transitions, has maximum 88.

$n = 14$. A 103-cell board would have wastage 36; all 261,225 physical low-charge depth-four belts and their centre completions give 224 stable 103-cell boards, none connected.

$n = 16, 17$. A corrected depth-six charge with the wastage congruence gives $C(16) \le 134$; at $n = 17$ the depth-five equality support for wastage 38 has a safe depth-six lift of minimum cost 39, excluding 152.

$n = 18, 19, 22$. Here $U(n)$ is one cell too weak. For the target population $P = U(n)$ the wastage is $W = 2n^2+4n-4P$, and the complete set of boundary rings of charge at most W is lifted inward, layer by layer, keeping every physical corner and extension that could still spend the remaining slack. The set becomes empty at depth seven for $n = 18$ ($W = 36$), at depth nine for $n = 19$ ($W = 42$), and at depth eleven for $n = 22$ ($W = 48$), where the straight sides have length zero. Verified boards of populations 170, 188, and 251 meet the resulting bounds. The 170-cell board at $n = 18$ does not come from component repair of the unrestricted optimum, where all 170-cell repairs have three components; it was recovered from the complete low-slack boundary language, and it is checked directly.

5 Bounds through fifty

Every board in the ledger was found by exact component repair of unrestricted optima, large-neighbourhood optimization, local replacement rules, or the periodic families; these methods are described in appendix D. Each board is checked

Table 2: Rigorous connected bounds for $18 \le n \le 50$: L_n is the verified construction, $M(n)$ the unrestricted optimum, $U(n)$ the bound of theorem 3.4, and $\overline{C}(n)$ the final upper bound. The dagger marks the specialized exclusions at $n = 18, 19, 22$. At $n = 18, 20, 23, 28, 31, 36, 39, 44, 47$ the unrestricted optimum satisfies $M(n) \le U(n)$; elsewhere the boundary bound is strictly stronger.

n	L_n	$M(n)$	$U(n)$	$\overline{C}(n)$	gap	n	L_n	$M(n)$	$U(n)$	$\overline{C}(n)$	gap
18	170	171	171	$170^{\dagger}$	0	35	627	633	631	631	4
19	188	190	189	$188^{\dagger}$	0	36	665	668	668	668	3
20	210	210	210	210	0	37	704	706	704	704	0
21	230	232	230	230	0	38	738	744	742	742	4
22	251	253	252	$251^{\dagger}$	0	39	782	782	783	782	0
23	276	276	277	276	0	40	817	824	821	821	4
24	299	302	299	299	0	41	863	864	863	863	0
25	325	326	325	325	0	42	902	907	906	906	4
26	350	353	352	352	2	43	943	949	948	948	5
27	375	379	378	378	3	44	989	993	993	993	4
28	406	407	407	407	1	45	1037	1039	1037	1037	0
29	435	437	435	435	0	46	1078	1085	1083	1083	5
30	462	467	465	465	3	47	1131	1132	1132	1132	1
31	497	497	498	497	0	48	1174	1181	1178	1178	4
32	528	531	528	528	0	49	1228	1229	1228	1228	0
33	562	563	562	562	0	50	1276	1280	1279	1279	3
34	594	598	597	597	3						

on the complete exterior ring and for a single Moore component. Let L_n be the population of the retained board and $\overline{C}(n)$ the strongest upper bound obtained from $M(n)$, theorem 3.4, and the specialized closures. Table 2 gives the ledger from $n = 18$ to $n = 50$; seventeen of these sizes are exact and sixteen are open. The retained boards exceed the capped-brick constructions of appendix E at every size $22 \le n \le 50$ and the band families of section 6 at 20 of the 27 sizes $24 \le n \le 50$, equalling them at the other seven.

Small gaps do not by themselves indicate that a uniform formula is near. At $n = 28$ and $n = 47$ the intervals have width one, and in both cases $M(n) = U(n)$, so a further cell needs either a connected board that is an unrestricted optimum or an exclusion beyond the present boundary language. The construction-side difficulty is the compatibility of a dense periodic bulk with four finite caps and one global component; the upper-bound difficulty is that a shallow boundary relaxation need not detect a connectivity obstruction that occurs only after several inward layers.

6 Period-eight band families and a uniform lower bound

Let S be a binary $n_0 \times n_0$ seed and let $h \ge 2$ satisfy $n_0 - h - 8 \ge 2$. Call rows $h, \dots, h+7$ and columns $h, \dots, h+7$ of S the *bands*. For $k \ge 1$ put $n = n_0 + 8(k-1)$ and

$$f_k(i) = \begin{cases} i, & 0 \le i < h, \\ h + ((i-h) \bmod 8), & h \le i < h + 8k, \\ i - 8(k-1), & h + 8k \le i < n, \end{cases} \qquad G_k(i,j) = S\big(f_k(i), f_k(j)\big). \tag{9}$$

Thus $G_1 = S$, and G_{k+1} arises from G_k by inserting one more copy of each band. The rows of G_k fall into a top cap of h rows, k copies of the band, and a bottom cap, and likewise the columns. The resulting nine tile types are four corners, four sides, and one centre, and the content of a tile depends only on its type. If A, B, C count the live cells in the central tile, in the four side tiles together, and in the four corner tiles together, then

$$|G_k| = Ak^2 + Bk + C \qquad (k \ge 1). \tag{10}$$

Nine seeds were found by the search methods of appendix D; they are stored in `ancillary/data/families.json`. For each residue $r \in \{0, \dots, 7\}$ there is a seed of side $24 + r$, and a ninth seed of side 40 improves residue zero. In every family the central 8×8 tile has half density, $A = 32$ (table 3).

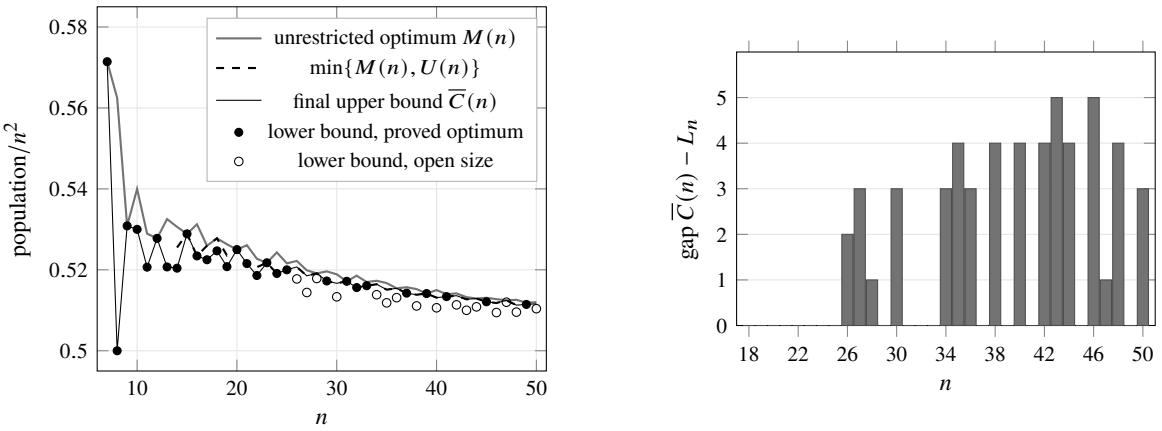


Figure 4: Left: unrestricted optimum, connected upper bound, and verified lower bound for $7 \le n \le 50$, normalized by area; filled markers are proved optima. Densities can exceed $1/2$ because boundary cells receive the $O(n)$ edge bonus of (5). Right: the remaining gap $\overline{C}(n) - L_n$, at most five, attained at $n = 43$ and $n = 46$.

Table 3: The nine band families: seed side n_0, band start h, tile counts A, B, C of (10), deficit $a_r = n(n+1)/2 - |G_k|$, and the radius R of the connectivity certificate. The primed row is the ninth family.

r	n_0	h	A	B	C	a_r	R
0	24	8	32	132	132	4	2
1	25	6	32	140	150	3	1
2	26	8	32	148	167	4	2
3	27	6	32	156	187	3	1
4	28	6	32	164	206	4	1
5	29	6	32	172	230	1	1
6	30	6	32	180	250	3	1
7	31	6	32	188	274	2	1
$0'$	40	16	32	260	525	3	1

Theorem 6.1 (Band families)**.** *Let $a = (a_0, \dots, a_7) = (4, 3, 4, 3, 4, 1, 3, 2)$. For every $n \ge 24$ there is a connected still life in $\mathcal{B}_n$ of population*

$$\frac{n(n+1)}{2} - a_{n \bmod 8}, \tag{11}$$

and for $n \equiv 0 \pmod 8$, $n \ge 40$, one of population $n(n+1)/2 - 3$. Consequently

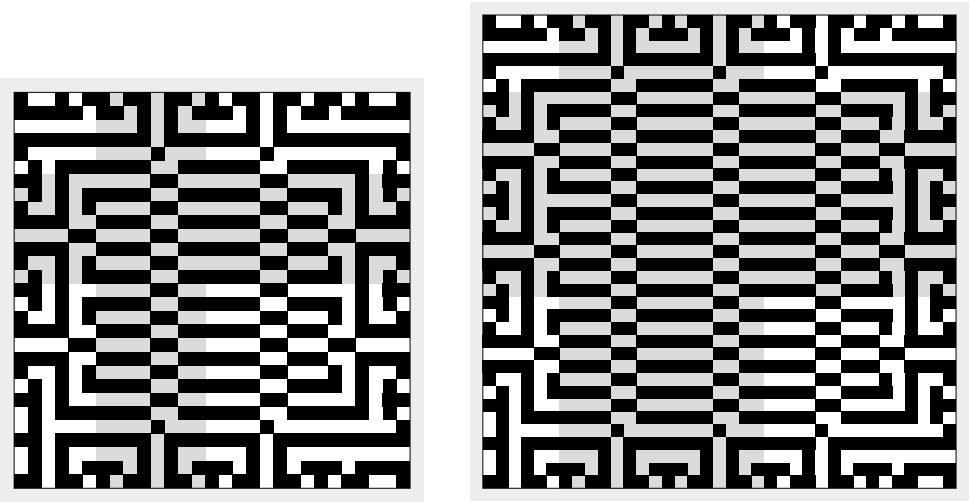
$$C(n) \ge \frac{n(n+1)}{2} - 4 \qquad \text{for every } n \ge 1, \tag{12}$$

and the constant 4 is best possible.

Stability is local, but connectivity is not. For each tile type list its Moore components and, for every pair of neighbouring tile types (cardinal or diagonal), the pairs of components joined across the common boundary. In each tile type call its largest component the *main* component. The tile grid of G_k has $k+2$ rows and columns, labelled top, band, …, band, bottom. For $R \ge 1$, the *context* of a tile position is the array of labels of the tiles within distance R in each coordinate, clipped at the boundary. A family is *R-linked* if, for every $k \in \{1, \dots, 2R+1\}$ and every tile position of G_k, inside the graph spanned by the tiles of its context, every component of the tile and the main component of each cardinally adjacent tile are joined to the main component of the tile.

Lemma 6.2 (Connectivity pumping)**.** *If a band family is R-linked, then G_k is connected for every $k \ge 1$.*

Proof. In one direction the clipped context at tile index $p \in \{0, \dots, k+1\}$ is determined by $(\min(p, R), \min(k+1-p, R))$. For $k \ge 2R+1$ the set of these pairs is $\{(p, R) : p \le R\} \cup \{(R, R)\} \cup \{(R, q) : q \le R\}$, independent of k



$k = 1$, $n = 29$, population 434 $k = 2$, $n = 37$, population 702

Figure 5: The band family of residue $r = 5$ at $k = 1$ (the seed, $n_0 = 29$) and $k = 2$ ($n = 37$). Shaded rows and columns are the repeated bands.

and already realized at $k = 2R + 1$. A context in the plane is a pair of a row and a column context, and tile contents and interfaces depend only on tile types, so the linking property holds at every position of every G_k. Hence every component is joined to the main component of its tile, the main components of cardinally adjacent tiles are joined, and since the tile grid is connected, so is G_k. □

Proof of theorem 6.1. Population. For each seed, table 3 gives $A = 32$, and substituting $k = (n - n_0)/8 + 1$ into (10) gives exactly $n(n + 1)/2 - a_r$. For instance, for $r = 5$ and $n = 21 + 8k$ one has $n(n + 1)/2 = 32k^2 + 172k + 231$ and $|G_k| = 32k^2 + 172k + 230$. Both sides are quadratics in k with leading coefficient 32, so comparing the other two coefficients suffices. The ninth family has $|G_k| = 32k^2 + 260k + 525 = n(n + 1)/2 - 3$ with $n = 32 + 8k$.

Stability. Stability is a condition on the 3×3 windows of the zero-extended array at the centres of $\mathcal{B}_n \cup R_n$. The caps and the band each have at least two lines, so three consecutive lines of G_k are three consecutive lines of S, or straddle a seam between two band copies (lines $h + 6, h + 7, h$ or $h + 7, h, h + 1$), possibly with dead lines beyond the boundary. The set of such types is the same for all $k \geq 2$, and a window is a pair of a row type and a column type, so every window of G_k with $k \geq 3$ occurs in G_2. Direct verification of G_1 and G_2 on $\mathcal{B}_n \cup R_n$ therefore proves stability for all k.

Connectivity. The base families with $r = 0$ and $r = 2$ are 2-linked, with 79 distinct contexts, and the other seven are 1-linked, with 30. Lemma 6.2 gives connectivity for every k.

The uniform bound. Every $n \geq 24$ equals $n_0 + 8(k - 1)$ for the seed with $n_0 = 24 + (n \bmod 8)$ and some $k \geq 1$, and the largest deficit is 4. For $n \leq 23$ the exact values give $C(n) - n(n + 1)/2 \geq -4$:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
$C(n) - \frac{n(n+1)}{2}$	−1	1	0	−2	−4	−3	0	−4	−2	−2	−3	−2	−3	−3	−1	−2	−2	−1	−2	0	−1	−2	0

Equality occurs at $n = 5$ and $n = 8$, so the constant 4 is best possible. The population formulas, the window containment for $3 \leq k \leq 8$, every context, and the absence of new contexts for $2R + 1 \leq k \leq 2R + 4$ are rechecked by the audit script of appendix F. □

Theorem 6.3 (Density)**.** $\lim_{n\to\infty} C(n)/n^2 = 1/2$. *More precisely,*

$$\frac{1}{2} \leq \liminf_{n\to\infty} \frac{C(n) - n^2/2}{n} \leq \limsup_{n\to\infty} \frac{C(n) - n^2/2}{n} \leq \frac{5}{8}.$$

Proof. The lower bounds follow from theorem 6.1, since $C(n) \geq n^2/2 + n/2 - 4$. The upper bounds follow from $C(n) \leq U(n) \leq n^2/2 + 5n/8$. □

The earlier families of appendix E are weaker: the pond lattice has density $4/9$, the capped bricks have density $1/2$ with a negative linear coefficient, and the period-fifty-four family has linear coefficient $13/27 < 1/2$. The bound (12) is not an upper bound either: $C(65) = 2150 = 65 \cdot 66/2 + 5$. Whether $C(n) - n(n+1)/2$ is bounded is open. The density statement is consistent with Elkies's theorem for infinite still lifes [1].

7 Open questions

The immediate finite problems are $350 \le C(26) \le 352$ and the one-cell intervals $406 \le C(28) \le 407$ and $1131 \le C(47) \le 1132$.

The chains of section 4.1 suggest a sharper question: is $C(n) = U(n)$ for every $n \equiv 5 \pmod 8$ with $n \ge 21$ and every $n \equiv 1 \pmod 8$ with $n \ge 25$? In the other classes the optimum equals $U(n)$ at $n = 16, 24, 32$ and is strictly smaller at $n = 14, 17, 18, 19, 22$.

The upper expression satisfies the exact period of corollary 3.5, but that is not a recurrence for $C(n)$: a construction pump must carry compatible states through all four corners, fill the enlarged interior stably, and preserve one component. The band families achieve this with linear coefficient $1/2$, while U has linear coefficient $5/8$, and the true linear coefficient is not known. A natural route to an exact formula is a pair of finite transducers whose states record the four boundary phases together with a connectivity signature of the interior interface; the present boundary automata control wastage but forget that global information.

The sequence $C(n)$ appears in the On-Line Encyclopedia of Integer Sequences as A398723 [7], with the unrestricted sequence A055397 as its companion upper bound [6]. All boards, certificates, scripts, and verifiers are available in the public repository [8].

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AI declaration

OpenAI's ChatGPT (Sol 5.6 and 6 Astra) was used mainly to find and run the search programs, certificate generators, and proofs, and Anthropic's Claude was used to help prepare this manuscript. The author directed the work, checked the claims, and is responsible for every statement. Every result rests on a verified board, a finite certificate, a cited theorem, or an argument given in the paper.

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A The CAISc 2026 challenge

The first Conference for AI Scientists (CAISc 2026) was held online on 24–25 July 2026; its Verifiable Problems Track paired each submission with an automatically checkable artifact [9]. For a prescribed size n, an artifact was a binary array $x \in \{0,1\}^{n\times n}$ and a claimed population p, accepted exactly when $p = \sum x_{ij}$, the B3/S23 conditions held on the board and its complete dead ring, and the support was one Moore component. An accepted artifact therefore proves $C(n) \geq p$ but says nothing about $C(n) \leq p$. Table 4 summarizes the seven accepted submissions; the entries report their own claims.

Table 4: Accepted CAISc 2026 connected-still-life submissions.

submission	stated results	principal methods
Collier [10]	exact for $n \in \{8, 9, 10, 11, 13\}$; 209, 495, 526 at $n = 20, 31, 32$	wastage-aware CP-SAT with lazy connectivity, MaxSAT, Chuffed, an IPASIR-UP propagator, stripe DP, cross-engine validation
Jaswal [11]	exact for $n \in \{8, 9, 10, 11, 15\}$; 133, 208, 495, 524 at $n = 16, 20, 31, 32$	pseudo-Boolean model with strict-rank connectivity and VeriPB proof logging, CP-SAT, large-neighbourhood search; the 2×8 brick motif
Menta [12]	exact for $n \in \{8, 9, 10\}$; 133, 207, 485, 515 at $n = 16, 20, 31, 32$	complete CP-SAT with breadth-first connectivity, symmetry, parallel tempering, verifier-gated acceptance
Reddy [13]	exact for $n \in \{8, 9, 15\}$; 133, 207, 495, 525 at $n = 16, 20, 31, 32$	CP-SAT, strip DP, symmetry audits, exact re-solving of pockets cut from dense still lifes
Rushdi [14]	exact for $n \in \{8, 9, 10, 11, 15\}$; 132, 206, 479, 515 at $n = 16, 20, 31, 32$	parent-pointer spanning tree in CP-SAT, local-search polishing
Sodha [15]	exact for $n \in \{8, 9, 10, 11, 15\}$; 133, 206, 445, 463 at $n = 16, 20, 31, 32$	wastage-aware CP-SAT and lazy constraints, GPU annealing
Sridhar [16]	exact for $n \in \{8, 9, 10, 11, 15\}$; 131, 208, 483, 515 at $n = 16, 20, 31, 32$	SAT and MaxSAT encodings, static connectivity, row-frontier DP, seeds, annealing, repair

Every board from these submissions was passed through the independent full-ring verifier; solver labels and declared populations were not trusted. The present paper raises the values at $n = 16, 19, 20, 31, 32$ to 134, 188, 210, 497, 528 and closes all thirteen sizes. Several ideas were carried over. Cross-engine rechecking follows Collier; the separation of an optimizer transcript from a replayable proof object follows Jaswal and Sridhar; complete static connectivity encodings of the kind used by Menta and Rushdi are used for small threshold instances; Reddy's pocket repair becomes the exact large-neighbourhood repair of appendix D; and Sodha's cost-of-connectivity viewpoint and Collier's wastage-aware search motivated the block-free boundary relaxation of section 3. Each theorem in this paper nevertheless rests on its own certificate or finite argument.

B Boundary certificate data

Depth three. The unrestricted depth-three edge graph has 222 states and 787 arcs with minimum cycle mean $4/11$; removing block-creating arcs leaves 214 states and 771 arcs with minimum cycle mean $3/8$, certified by the integer

potential of section 3. If $f(L)$ is the minimum cost of a free-ended block-free path of length L, a phase-potential check gives $f(L+8) = f(L) + 3$ for $L \geq 4$. All 2^{16} binary 4×4 patches are examined for the corner relation: 784 are valid physical corners, giving 722 state mappings.

Depth four. 809 edge states and 3685 arcs. The complete 5×5 corner census has 14,225 valid physical patches, 9957 minimum-cost mappings, and 711 distinct output states, so the recurrence check uses a 711×809 restricted path matrix; at both recurrence phases all 470,400 finite entries increase by three and the infinite entries are unchanged.

Depth five. 3051 edge states and 17,649 arcs; the 6×6 corner census has 500,297 valid physical patches, 179,476 minimum-cost mappings, and 2706 output states. For a target cost K, every closed four-step walk of cost at most K contains an arc of cost at most $\lfloor K/4 \rfloor$; anchoring every such arc and computing exact two- and three-step return costs enumerates all low-cost rings. At $n = 21, 29, 37, 45, 53$ the depth-five minima are $43, 55, 67, 79, 91$, one more than the depth-four charges $42, 54, 66, 78, 90$.

Depth six, residue zero. At $n = 16, 24, 32, 40, 48$ depth five attains the old minimum. The equality rings lie in a support of 15 corner-output states, 29 side arcs, 14 four-cycle shapes, and 23 physical equality corner mappings. The lift charges only the genuinely new inward corner cell and keeps, for each of the 23 corners, every valid incoming and outgoing depth-five edge and both choices of the new halo bit. The lifted language has 266 corner sources, 10,996 depth-six edge states, and 55,422 arcs, contains no ring at the old target for $n = 16, 24, 32, 40$, and from the $n = 40$ phase to the $n = 48$ phase the restricted matrix increases entrywise by three.

C Details of the specialized closures

C.1 Threshold encodings for $n \leq 11$

For each $v \in \mathcal{B}_n$ let x_v be the live-cell variable. The stability encoding forbids every assignment to a centre and its in-box neighbours that violates B3/S23, and for each centre of the exterior ring every assignment with exactly three live in-box neighbours; a sequential counter imposes $\sum_v x_v \geq K$. Connectivity chooses a root, the first live cell in row-major order; every other live cell chooses a Moore-adjacent live parent u and every cell a binary rank $r_v \in \{0, \ldots, n^2 - 1\}$ with $p_{uv} = 1 \Rightarrow x_u = x_v = 1$ and $r_u < r_v$. A second component would contain a minimum-rank vertex without a parent of smaller rank, and conversely a connected live set admits such a tree, so the encoding is complete. Requiring a live cell on the top and on the left side removes translations; no reflection or rotation is removed. The excluded thresholds K are

n	2	3	4	5	6	7	8	9	10	11
construction	4	6	8	11	18	28	32	43	53	63
excluded K	5	7	9	12	19	29	33	44	54	64

Complete CNF and DRUP streams are retained and replayable through $n = 7$. The streams for $n = 8, 9, 10, 11$, of about 35 MB, 79 MB, 585 MB, and 1.44 GB, were independently replayed when generated; their hashes and replay records are retained.

C.2 The block-free row program at $n = 13$

Any connected candidate with at least 89 cells is block-free, so it suffices to optimize over all block-free stable boards. Rows are masks in $\{0,1\}^{13}$; for consecutive masks a, b, c let $T(a, b, c)$ hold when every cell of b satisfies the Life rule, the dead centres to the left and right of b are not born, and rows b, c contain no live block. With $F_k(a, b)$ the maximum population of a partial board with final row pair (a, b),

$$F_{k+1}(b, c) = \max_{a : T(a,b,c)} \big(F_k(a, b) + \mathrm{pop}(c)\big),$$

starting from the dead pair $(0, 0)$ and ending with two forced zero rows. Successors are generated left to right with pruning of invalid prefixes. The computation examines 3,926,925,326 transitions, with 24,235,166 active row-pair states after row 13 and final maximum 88. An exhaustive generator agrees with the pruned one on 21,844 row pairs and 63,318 successors; the same recurrence gives block-free maxima $0, 0, 6, 8, 12, 18, 28, 33$ for widths 1 to 8; and GCC and Clang reconstructions agree in layer counts, maxima, transition counts, and three 64-bit layer digests. Hence $C(13) = 88$.

C.3 Complete low-wastage closure at $n = 14$

The depth-four theorem gives $102 \le C(14) \le 103$, and a 103-cell board has $W = 2 \cdot 14^2 + 4 \cdot 14 - 4 \cdot 103 = 36$, so its depth-four belt has charge at most 36. The complete low-charge corner language has 2664 abstract four-cycle orbits modulo rotation; all tied and near-minimum physical patches within the slack are expanded, giving 72,193, 87,672, and 101,360 physical belts at baseline charges 34, 35, 36 (allowances 2, 1, 0), in total 261,225. Every one of the 2^{16} assignments of the remaining 4×4 centre is tested against each belt. The 224 stable 103-cell boards have component signatures $49 + 38 + 6 + 6 + 4$ (25), $54 + 39 + 6 + 4$ (60), $87 + 6 + 6 + 4$ (75), $91 + 6 + 6$ (8), $93 + 6 + 4$ (52), and $97 + 6$ (4); none is connected, so $C(14) = 102$.

C.4 Closures at $n = 16$ and $n = 17$

At $n = 16$ a board of population at least 135 has wastage at most 36. The corrected depth-six equality lift proves $B_6(16) > 36$, the minimum anchored depth-six ring having charge 38, and $W \equiv 2 \cdot 16^2 + 4 \cdot 16 \equiv 0 \pmod 4$ forces $W \ge 40$, so $|S| \le (576 - 40)/4 = 134$. At $n = 17$ a 152-cell board has $W = 38$, so its block-free depth-five projection is a minimum-charge equality ring. The equality support has 21 states and 50 arcs, with 27 physical equality patches, and the sixth layer must add zero. The safe depth-six lift retains every locally valid zero-added-cost continuation and has minimum four-side cost 39, so no 152-cell board exists and $C(17) = 151$.

C.5 The closure at $n = 18$

The unrestricted 18×18 optimum has population 171 and signature $159 + 4 + 4 + 4$. Coupled component-reducing repair gives 21 stable 170-cell states with three components, 128 with population 169 and two components, and 240 connected 168-cell states; freeing the 144 cells within radius four of the three blocks gives exact restricted optimum 168. A connected 171-cell board would have total wastage 36. The complete depth-five equality support at this target has four states and eight arcs, its zero-added-wastage depth-six support has four states and four self-loops, and the safe depth-seven target-36 support is empty; so $C(18) \le 170$. The verified 170-cell board was recovered from the complete inward low-slack boundary language: at terminal depth nine, where the straight sides of the 18×18 board have length zero, a four-corner assembly covers the whole board.

C.6 The closure at $n = 19$

The unrestricted 19×19 optimum has population 190 and signature $142 + 12 \cdot 4$, and $U(19) = 189$. A population-189 board has $W = 42$, while the exact depth-five minimum is 40, so physical realizations that spend the two remaining units must be kept. Let $T_d(i, j)$ be the minimum charge of a depth-d side followed by a corner from state i to state j, M_d the minimum cost of a closed four-step walk in T_d, and $S_d = 42 - M_d$. For every arc in a closed walk of cost at most 42, the lifting keeps every physical corner realization of charge at most $T_d(i, j) + S_d$, and then lifts every next-layer edge and corner of added charge at most S_d. Replacing each realization in the projection of a population-189 board by its abstract minimum gives a supported cycle of cost at least M_d, so the total excess is at most S_d and every such projection survives: the language is a safe overapproximation (table 5).

Table 5: Low-slack lifting for population 189 at $n = 19$.

depth	straight length	minimum	slack	physical candidates	retained corners	target support
5	9	40	2	500,297	133	38 states, 123 arcs
6	7	42	0	9,785	19	11 states, 18 arcs
7	5	42	0	6,610	4	4 states, 4 arcs
8	3	42	0	1,252	4	4 states, 4 arcs
9	1	none	none	2,328	0	empty

At depth nine there are two finite arcs of charge at most $\lfloor 42/4 \rfloor = 10$ and neither has a finite three-step return, while every closed four-step walk of charge at most 42 contains such an arc. So population 189 is impossible and $C(19) = 188$.

C.7 The closure at $n = 22$

Here $M(22) = 253$ and $U(22) = 252$, and a 252-cell board has $W = 48$. The same lifting with target 48 runs from depth five, where the minimum is 45 and the slack three, through depth six onward, where the minimum is 48 and the slack zero, to the terminal depth eleven, where the straight sides have length zero. The numbers of retained physical corner classes at depths $5, \dots, 11$ are $1309, 62, 104, 112, 42, 4, 0$. At depth eleven a side transfer is the identity join, so consecutive corners can be joined only if the output state of one equals the incoming state of the next, and among the 2592 minimum corner relations, with 180 distinct output states, no compatible pair exists. The earlier layers are essential; the terminal incompatibility alone would not suffice. Hence $C(22) = 251$.

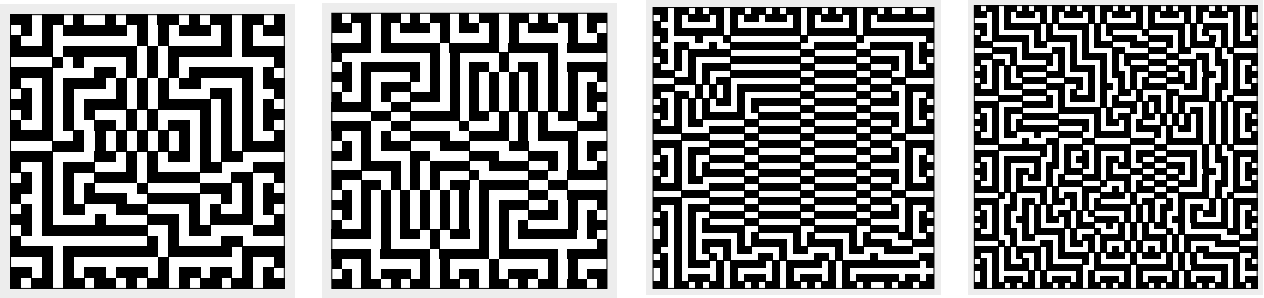
D Construction and repair methods

Every lower bound is a complete binary board; the eight symmetries of the square are factored out by keeping the lexicographically least representative.

Exact large-neighbourhood repair. Let $Y \subseteq \mathcal{B}_n$ be a stable, possibly disconnected source and $R \subseteq \mathcal{B}_n$ the cells allowed to change, and let R^+ be the cells of $\mathcal{B}_n \cup R_n$ within Chebyshev distance one of R. Fixing $x_z = \mathbf{1}_Y(z)$ outside R and imposing the Life rule at every centre of R^+ describes exactly the stable boards that agree with Y outside R. Population is maximized and connectivity imposed by a rooted parent-and-rank system or by an exact acceptance gate. Useful regions are neighbourhoods of small components, corridors joining them to the main component, side strips, quadrants, and unions of overlapping neighbourhoods. Disjoint defects may be repaired in sequence, while overlapping defects must be optimized together, since absorbing one block can create a birth or saturate a cell near another.

A replacement grammar. Each successful repair gives a pair of local patterns with a fixed radius-two halo, usable wherever its halo agrees; merging rules gives 72 component-reducing rules from 145 change clusters. Exhausting this grammar is a statement about the rule set, not a global upper bound; it produces lower bounds and selects regions for exact optimization.

A representative repair at $n = 40$. An unrestricted 40×40 optimum has population 824 and component signature $776 + 12 \cdot 4$. Sequential one-cell repairs give a connected board of 812 cells, a coupled radius-four optimization gives 814, and large-neighbourhood search gives 815. A later search reached 817, and that board is the seed of the ninth band family, so $817 \le C(40) \le 821$. The example shows why an unrestricted optimum is a good density seed but a poor connected construction: its excess population sits in components that cannot be joined without coordinated changes.



$n = 26, 350$ $n = 28, 406$ $n = 40, 817$ $n = 47, 1131$

Figure 6: Verified boards at four open sizes: the first open case $n = 26$, $n = 28$ one cell below the unrestricted optimum, the endpoint of the $n = 40$ search, and $n = 47$ with gap one.

E Earlier periodic families

A period-six pond lattice. Let $P = \{(0,1), (0,2), (1,0), (1,3), (2,0), (2,3), (3,1), (3,2)\}$, the eight-cell pond in a 4×4 square, $m = \lfloor (n-1)/3 \rfloor$, and $S_n^{\text{pond}} = \bigcup \{P + (3i, 3j) : 0 \le i, j < m,\ i \equiv j \pmod 2\}$. The copies are disjoint and lie in $\mathcal{B}_n$, diagonally consecutive ponds are Moore-adjacent, the index set is connected under diagonal steps, and the only

neighbourhoods are those of an isolated pond, a diagonal interface, and the outer boundary. Hence for $n \geq 4$, S_n^{pond} is a connected still life of population $8(\lceil m/2 \rceil^2 + \lfloor m/2 \rfloor^2) = (4/9)n^2 + O(n)$.

Period-eight capped bricks. The interior is the 2×8 tile

$$T = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

of Jaswal [11], repeated with periods two and eight. For each residue $r \pmod 8$ a fixed width-four template replaces the four strips and corners; a finite local certificate checks every 3×3 window at a cap, an interface, or a corner, and a finite quotient graph checks connectivity. This gives connected still lifes of population $P_r(n)$ for all $n \equiv r \pmod 8$ above the thresholds of table 6.

Table 6: Period-eight capped-brick populations.

r	least n	$P_r(n)$	r	least n	$P_r(n)$
0	16	$\frac{1}{2}n^2 - 3n + 13$	4	12	$\frac{1}{2}n^2 - \frac{19}{8}n + \frac{19}{2}$
1	17	$\frac{1}{2}n^2 - \frac{9}{2}n + 5$	5	13	$\frac{1}{2}n^2 - \frac{11}{8}n + \frac{51}{8}$
2	10	$\frac{1}{2}n^2 - \frac{5}{8}n - \frac{23}{4}$	6	14	$\frac{1}{2}n^2 - 3n + 12$
3	11	$\frac{1}{2}n^2 - \frac{23}{8}n - \frac{87}{8}$	7	15	$\frac{1}{2}n^2 - \frac{3}{4}n + \frac{19}{4}$

A period-fifty-four macrotile family. Split the rows of a connected 154×154 board into blocks of heights $42, 54, 58$ and its columns into blocks of widths $48, 54, 52$, and repeat the central row and column bands together k times ($k = 0$ deletes one copy of each). The side is $n = 100 + 54k$ and the nine macrotile populations give a connected still life of population $1458k^2 + 5426k + 5051 = n^2/2 + 26k + 51$. The members $k = 0, 1, 2$ contain every cap and seam type, so stability for all k follows from finitely many windows; the largest component of every macrotile meets the largest components across every interface, and the secondary components in the right-hand macrotiles have certified interface edges to adjacent main components.

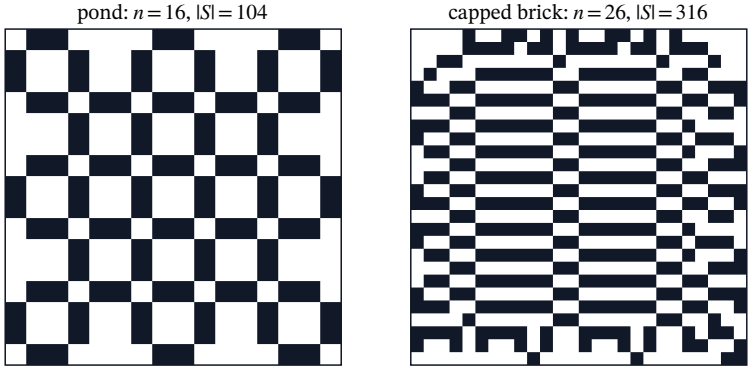


Figure 7: Members of the pond and capped-brick families.

F Computational certificates and ancillary data

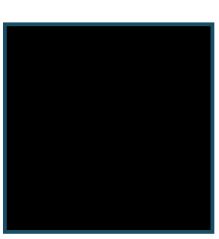
The public repository [8] contains the verified boards for $2 \leq n \leq 50$, the boundary, equality-support, closure, and row certificates, two independent board verifiers, regeneration scripts, regression tests, and a proof index recording the upper-bound route for each exact value of its ledger. The certificates for the small proof-logged exclusions are those distributed with the first version of this paper. The directory `ancillary/` accompanying this version contains what is new in it: the eleven boards that improve lower endpoints at $n = 26, 29, 35, 37, 40, 41, 43, 45, 46, 47, 49$ (closing $n = 29, 37, 41, 45, 49$), the boards at $n = 53, 55, 57, 65$, the nine band-family seeds, the updated ledger, the script that regenerates the figures and tables, and a standalone audit script.

The construction verifiers recompute the Life rule and connectivity from the grid and do not trust stored counts. The SAT objects are threshold statements: a satisfying assignment at the target would refute the bound, whereas a checked refutation proves it.

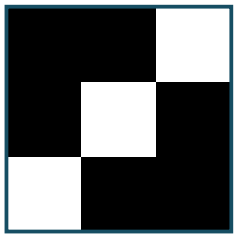
Running `python3 ancillary/scripts/verify_manuscript.py`, which uses only the Python standard library, recomputes the population, stability on the full dead ring, and connectivity of every board used here; rebuilds the ledger for $2 \le n \le 50$; checks the statements about exact values, open intervals, and the chains of section 4.1; and certifies the nine band families as in the proof of theorem 6.1. It reads the upper bounds of the repository ledger and does not replay the boundary, row, or closure certificates, which are audited by the repository's own scripts. The unrestricted maxima $M(n)$ are cited, not recomputed. No proof-assistant formalization is claimed.

G Atlas of verified boards

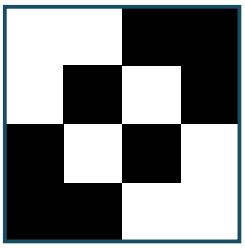
The verified board for every n from 2 to 50, and the optimal boards at $n = 53, 57, 65$. Solid blue frames mark proved optima; dashed gray frames mark open sizes, labelled with the current interval.



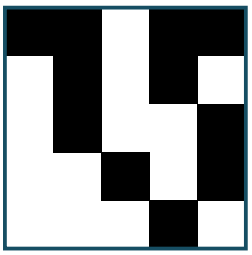
$n = 2$: $C(n) = 4$



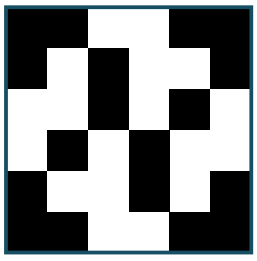
$n = 3$: $C(n) = 6$



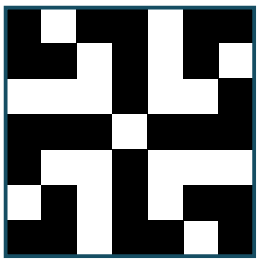
$n = 4$: $C(n) = 8$



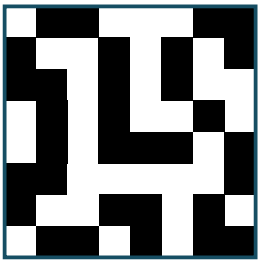
$n = 5$: $C(n) = 11$



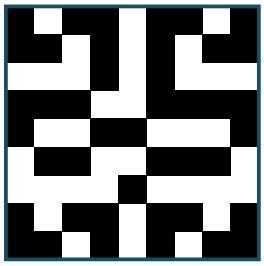
$n = 6$: $C(n) = 18$



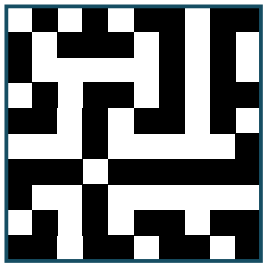
$n = 7$: $C(n) = 28$



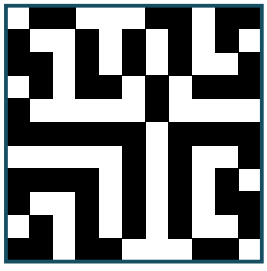
$n = 8$: $C(n) = 32$



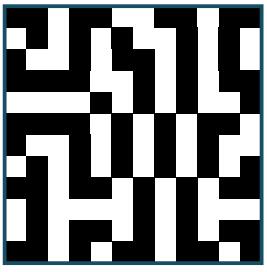
$n = 9$: $C(n) = 43$



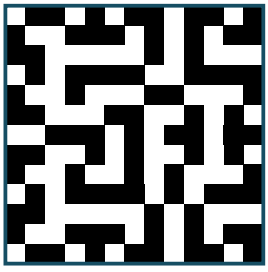
$n = 10$: $C(n) = 53$



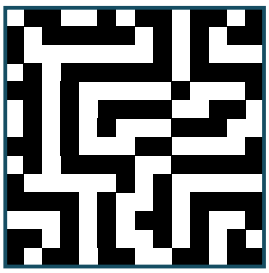
$n = 11$: $C(n) = 63$



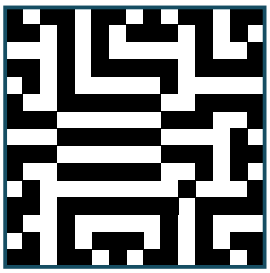
$n = 12$: $C(n) = 76$



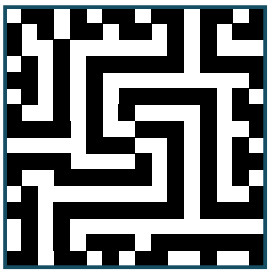
$n = 13$: $C(n) = 88$



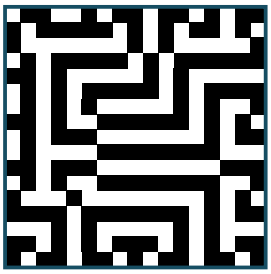
$n = 14$: $C(n) = 102$



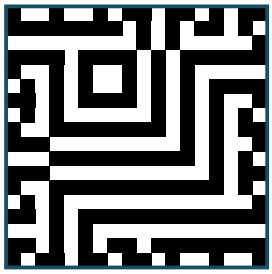
$n = 15$: $C(n) = 119$



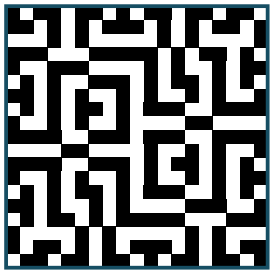
$n = 16$: $C(n) = 134$



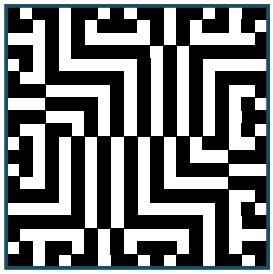
$n = 17$: $C(n) = 151$



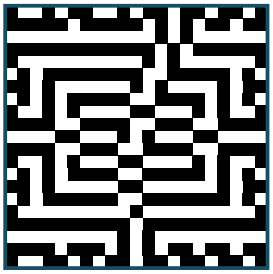
$n = 18$: $C(n) = 170$



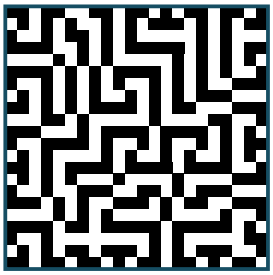
$n = 19$: $C(n) = 188$



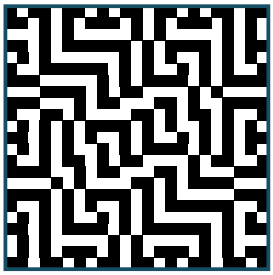
$n = 20$: $C(n) = 210$



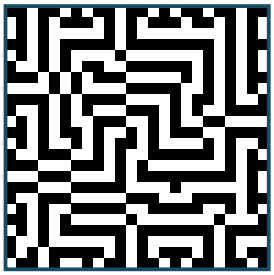
$n = 21$: $C(n) = 230$



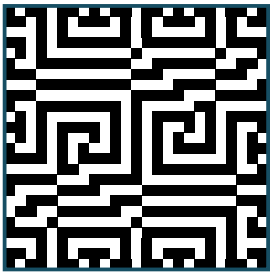
$n = 22$: $C(n) = 251$



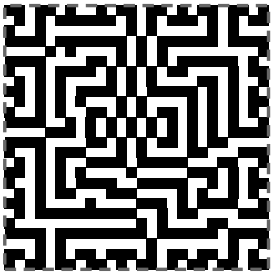
$n = 23$: $C(n) = 276$



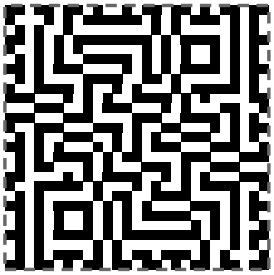
$n = 24$: $C(n) = 299$



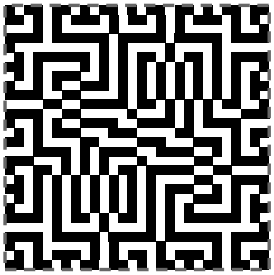
$n = 25$: $C(n) = 325$



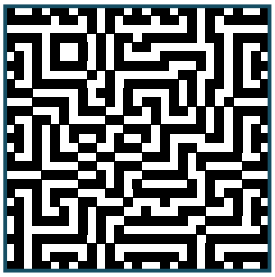
$n = 26$: $350 \leq C(n) \leq 352$



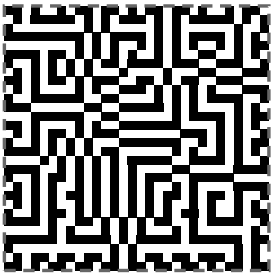
$n = 27$: $375 \leq C(n) \leq 378$



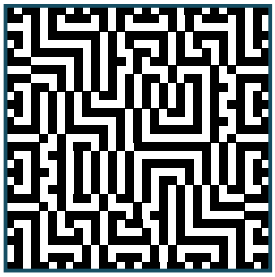
$n = 28$: $406 \leq C(n) \leq 407$



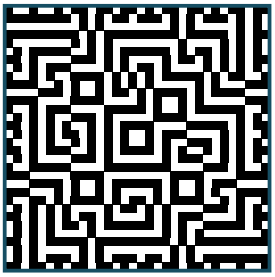
$n = 29$: $C(n) = 435$



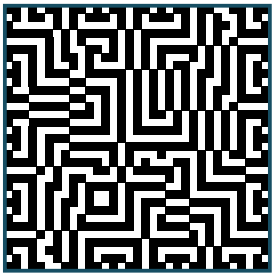
$n = 30$: $462 \leq C(n) \leq 465$



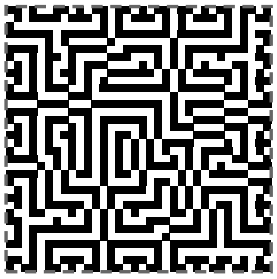
$n = 31$: $C(n) = 497$



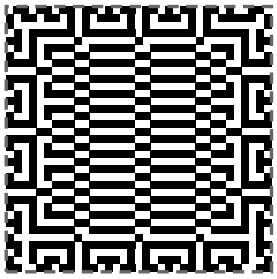
$n = 32$: $C(n) = 528$



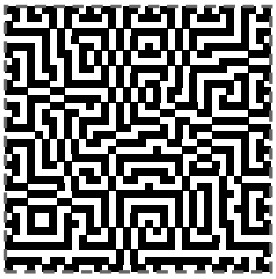
$n = 33$: $C(n) = 562$



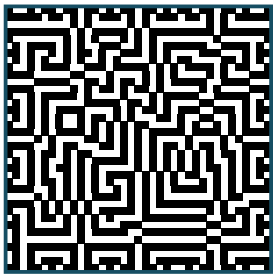
$n = 34$: $594 \leq C(n) \leq 597$



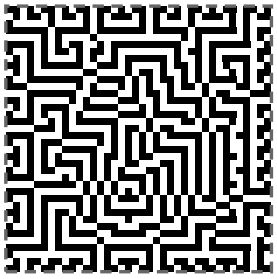
$n = 35$: $627 \leq C(n) \leq 631$



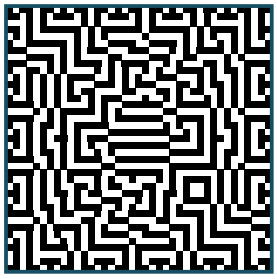
$n = 36$: $665 \leq C(n) \leq 668$



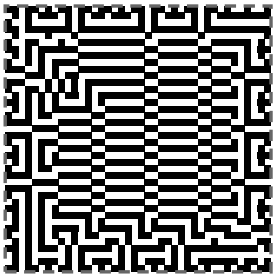
$n = 37$: $C(n) = 704$



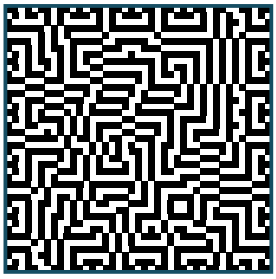
$n = 38$: $738 \leq C(n) \leq 742$



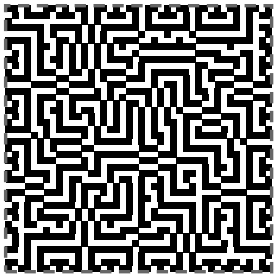
$n = 39$: $C(n) = 782$



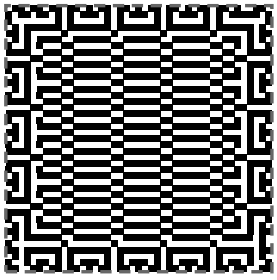
$n = 40$: $817 \leq C(n) \leq 821$



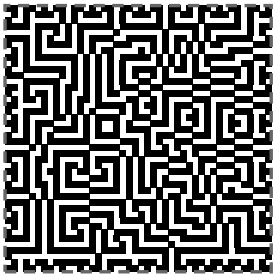
$n = 41$: $C(n) = 863$



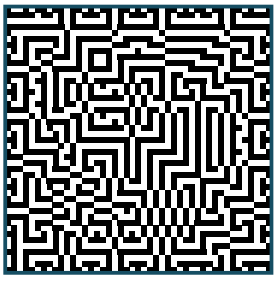
$n = 42$: $902 \leq C(n) \leq 906$



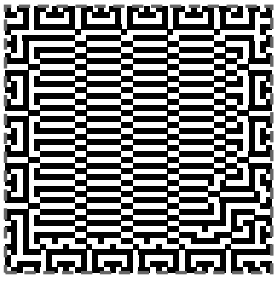
$n = 43$: $943 \leq C(n) \leq 948$



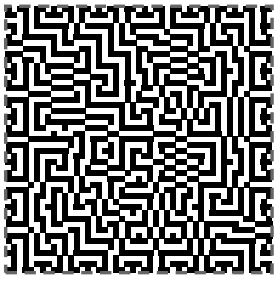
$n = 44$: $989 \leq C(n) \leq 993$



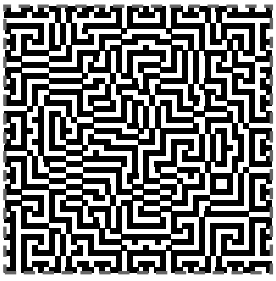
$n = 45$: $C(n) = 1037$



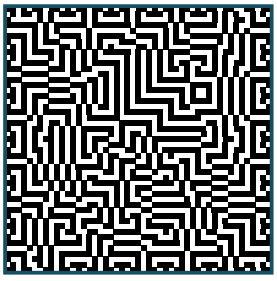
$n = 46$: $1078 \leq C(n) \leq 1083$



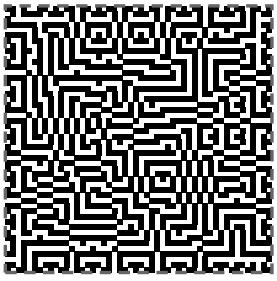
$n = 47$: $1131 \leq C(n) \leq 1132$



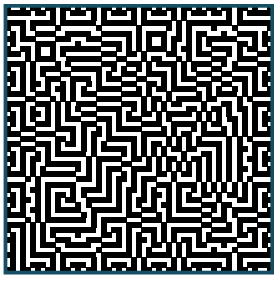
$n = 48$: $1174 \leq C(n) \leq 1178$



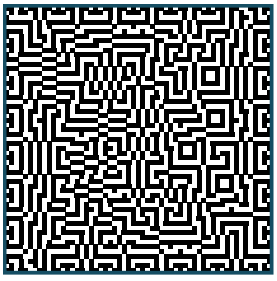
$n = 49$: $C(n) = 1228$



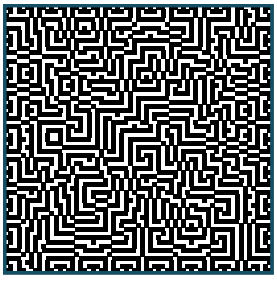
$n = 50$: $1276 \leq C(n) \leq 1279$



$n = 53$: $C(n) = 1434$



$n = 57$: $C(n) = 1657$



$n = 65$: $C(n) = 2150$